ME 264 Mechanisms Synthesis & Analysis I

Unit 4 Acceleration Analysis

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From Velocity Analysis

$$r_B = r_A + r_{B/A} \qquad (1)$$

- Where $r_{B/A}$ is the position of point B relative to point A. Let

$$r_{B/A} = xi + yj + zk \qquad (2)$$

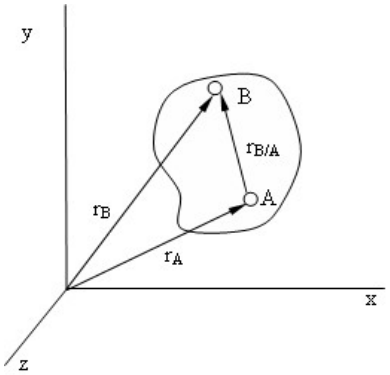
Differentiating Equation (1) with respect to time gives

$$v_B = v_A + \left(\frac{dx}{dt}i + x\frac{di}{dt}\right) + \left(\frac{dy}{dt}j + y\frac{dj}{dt}\right) + \left(\frac{dz}{dt}k + z\frac{dk}{dt}\right)$$

$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \left(x\frac{di}{dt} + y\frac{dj}{dt} + z\frac{dk}{dt}\right) \qquad (3)$$

Differentiating the unit vectors with respect to time only causes rotation of the vectors. Thus,

$$\frac{di}{dt} = w \times i \qquad \frac{dj}{dt} = w \times j \qquad \frac{dk}{dt} = w \times k \qquad (4)$$



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Summary of Velocity Analysis

$$r_B = r_A + r_{B/A} \quad (1)$$

- Where $r_{B/A}$ is the position of point B relative to point A. Let

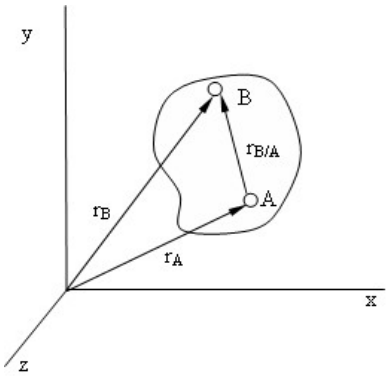
$$r_{B/A} = xi + yj + zk \quad (2)$$

Differentiating Equation (1) with respect to time gives

$$v_B = v_A + v_{B/A} + w \times r_{B/A} \quad (3)$$

where $v_{B/A} = \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right)$

v_A and v_B are respectively velocities of point A and B relative to the reference frame, and $V_{B/A}$ is the sliding velocity of point B relative to point A



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Acceleration Equation

$$a_B = a_A + \frac{d}{dt}v_{B/A} + \frac{d}{dt}(w \times r_{B/A}) \quad \text{----------(1)}$$

$$\frac{d}{dt}v_{B/A} = \frac{d}{dt}\left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right)$$

$$\frac{d}{dt}v_{B/A} = \left(\frac{d^2x}{dt^2}i + \frac{d^2y}{dt^2}j + \frac{d^2z}{dt^2}k\right) + \omega \times \left[\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right] \quad \text{----------(2)}$$

$$\frac{d}{dt}v_{B/A} = a_{B/A} + w \times V_{B/A}$$

$$\frac{d}{dt}\left(w \times r_{B/A}\right) = \frac{d}{dt}w \times r_{B/A} + w \times \frac{d}{dt}r_{B/A}$$

$$\frac{d}{dt}\left(w \times r_{B/A}\right) = \alpha \times r_{B/A} + w \times \left(v_{B/A} + w \times r_{B/A}\right) \quad \text{----------(3)}$$

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Acceleration Equation

Differentiating Equation (3), the velocity equation, with respect to time gives

$$a_B = a_A + a_{B/A} + 2\omega \text{ x } v_{B/A} + \alpha \text{ x } r_{B/A} + \omega \text{ x } (\omega \text{ x } r_{B/A}) \quad (4)$$

Spatial (3D) motion

$$a_B = a_A + a_{B/A} + 2\omega \text{ x } v_{B/A} + \alpha \text{ x } r_{B/A} - \omega^2 \, r_{B/A} \quad (5)$$

Planer (2D) motion

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Case Study using Acceleration equation

- Case I: Pure Translational Motion

  $a_{B/A}=0$, $\alpha=0$ Therefore, Equation (5) reduce to $a_B = a_A$

- Case II: General Motion without Sliding on Rotating Link

  $a_{B/A}=0$ Therefore, Equation (5) reduce to $a_B = a_A + \alpha \text{ x } r_{B/A} - \omega^2 \, r_{B/A}$

- Case III: General Motion with Sliding on rotating Link

$$a_B = a_A + a_{B/A} + 2\omega \text{ x } v_{B/A} + \alpha \text{ x } r_{B/A} - \omega^2 \, r_{B/A}$$

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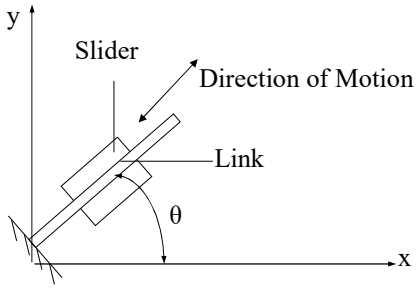
Acceleration Analysis using the Vector Method Cases

- Case IV: General Motion with Sliding on Fixed Link (also applicable to cases I, II & III if there is sliding)
  - Consider a body sliding on a link that is inclined at θ to the horizontal in x-y plane.
  - If the magnitude of the sliding velocity is A, then

    $A_{slider} = A(\cos\theta \boldsymbol{i} + \sin\theta \boldsymbol{j})$
  - $a_{slider} = a_A$ if the link is fixed
  - $a_{slider} = a_{B/A}$ if the link is rotating

    For horizontal motion, $A_{slider} = A\boldsymbol{i}$

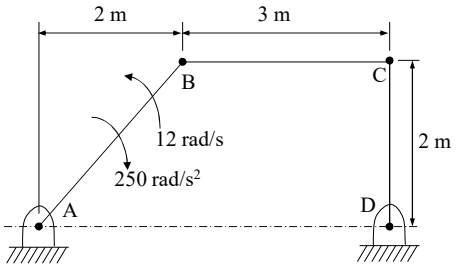
    For vertical motion, $A_{slider} = A\boldsymbol{j}$



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Example 4.1

Link AB shown has an anti-clockwise angular velocity of 12 rad/s and a clockwise angular acceleration of 250 rad/s^2. Determine the angular accelerations of links BC and CD.



Solution

Since the angular acceleration formula has velocity components, first the velocities of the links BC and CD are determined. Then, the determined velocities are used to determine accelerations.

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Velocity Analysis

Then, the determined velocities are used to determine accelerations of the links. The velocity of point B

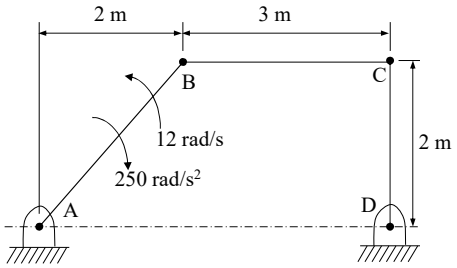
$$v_B = v_A + \omega_{AB} \times r_{B/A} \qquad v_B = 0 + 12\,k \times (2i + 2j)$$

$$v_B = -24i + 24j \quad \text{(m/s)}$$

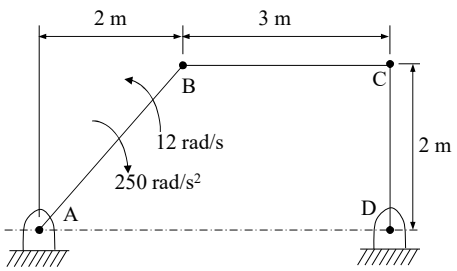
Let ω_{BC} and ω_{CD} the angular velocities of links BC and CD, respectively. The velocity of point C in terms of the velocity of point B is

$$v_C = v_B + \omega_{BC} \times r_{C/B} \qquad v_C = (-24i + 24j) + (\omega_{BC}k) \times (3i)$$

$$v_C = -24i + (3\omega_{BC} + 24)j \quad \text{----------(1)}$$



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Similarly, the velocity of point C in terms of the velocity of point D is

$$v_C = v_D + \omega_{CD} \times r_{C/D}$$

$$v_C = 0 + (\omega_{CD}k) \times (2j) = -2\omega_{CD}i \quad \text{----------(2)}$$

Equating the two expressions for the velocity of point C, i.e. equations (1) and (2), yields

$$-24i + (3\omega_{BC} + 24)j = -2\omega_{CD}i \quad \text{----------(3)}$$

Equation the i and the j components of the above equation and solving for the angular velocities, we have

For i: $-2\omega_{CD} = -24$ $\qquad \omega_{CD} = 12$ rad/s

For j: $(3\omega_{BC} + 24) = 0$ $\qquad \omega_{BC} = -8$ rad/s

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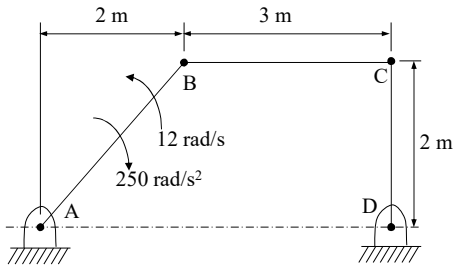
Acceleration Analysis

The acceleration of point B in terms of that of point A is

$$a_B = a_A + \alpha \text{ x } r_{B/A} - \omega^2 r_{B/A}$$

$$a_B = 0 - 250\text{k x}(2\text{i} + 2\text{j}) - (12^2)(2i + 2j)$$

$$a_B = 212\,i - 788\,j$$



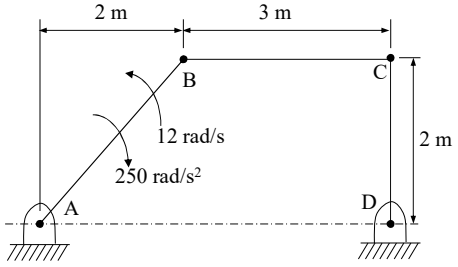
Let α_{BC} and α_{CD} the angular acceleration of links BC and CD, respectively. The acceleration of point C in terms of the acceleration of point B is

$$a_C = a_B + \alpha_{BC} \text{ x } r_{C/B} - \omega_{BC}^2 r_{C/B}$$

$$a_C = (212i - 788j) + \alpha_{BC}\text{k x } 3\text{i} - (-8)^2(3i)$$

$$a_C = 20i + (3\alpha_{BC} - 788)j \qquad \text{----------(4)}$$

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Similarly, the acceleration of point C in terms of the velocity of point D is

$$a_C = a_D + \alpha_{CD} \text{ x } r_{D/C} - \omega_{CD}^2 r_{C/D}$$

$$a_C = 0 + \alpha_{CD}\text{k x } 2\text{j} - (12)^2(2j)$$

$$a_C = -2\alpha_{CD}\text{i} - 288j \qquad \text{----------(5)}$$

Equating i and j components of Equations (4) and (5) and solving for the angular velocities, we have

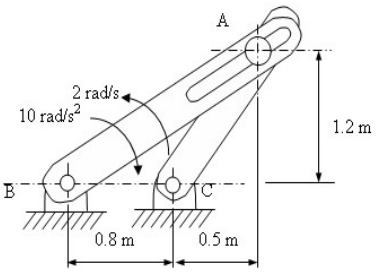
For i: $-2\alpha_{CD} = 20$ $\quad \alpha_{CD} = -10$ rad/s²

For j: $3\alpha_{BC} - 788 = -288$ $\quad \alpha_{BC} = 166.7$ rad/s²

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Example 4.2

For the mechanism shown, determine the angular acceleration of bar AC and acceleration of pin A relative to the slot in bar AB if bar AB has an anticlockwise velocity of 2 rad/s and a clockwise angular acceleration of 10 rad/s^2



Solution

Velocity of point A is given as

$$v_A = v_B + v_{A/B} + \omega_{BA} \text{ x } r_{A/B}$$

$$v_A = 0 + v_{A/B} + 2\text{k x}(1.3i + 1.2j) \quad (1)$$

$$v_A = v_{A/B} + (-2.4i + 2.6j) \quad (2)$$

$$v_{A/B} = |v_{A/B}|(\cos\theta i + \sin\theta j)$$

Where $\theta = \tan^{-1}\left(\frac{1.2}{1.3}\right) = 42.7$

Substituting for θ and (2) into (1) we have

$$v_A = (0.7384|v_{A/B}| - 2.4)i + (0.6783|v_{A/B}| + 2.6)j \quad (3)$$

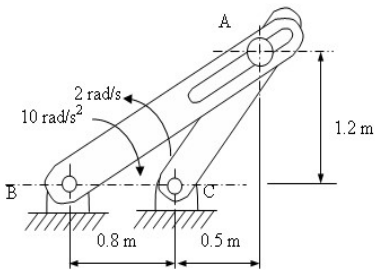
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Example 4.2

Let ω_{CA} and α_{CA} be the angular velocity and angular acceleration of link CA. The velocity of point A from C is

$$v_A = v_C + \omega_{CA} \text{ x } r_{A/C} \qquad v_A = 0 + \begin{bmatrix} i & j & k \\ 0 & 0 & \omega_{CA} \\ 0.5 & 1.2 & 0 \end{bmatrix}$$

$$v_A = -1.2\omega_{CA} i + 0.5\omega_{CA} j \quad (4)$$



Equating the two velocities of A gives

$$-1.2\omega_{CA} i + 0.5\omega_{CA} j = (0.7384 v_{Arel} - 2.4)i + (0.6783 v_{Arel} + 2.6) \quad (5)$$

Equating i and j components of the above equation yields

$$-1.2\omega_{CA} = (0.7384 v_{Arel} - 2.4) \quad (6)$$

$$0.5\omega_{CA} = (0.6783 v_{Arel} + 2.6) \quad (7)$$

Solving the above two equations, we have

$$\omega_{CA} = 2.995 \qquad v_{A/B} = -1.625$$

The velocity and acceleration of A relative B are resolved into i and j components as

$$v_{A/B} = -3.5258(\cos 42.709 i + \cos 42.709 j) = -2.5907i - 2.3915j$$

$$\vec{a}_{A/B} = \alpha_{A/B}(\cos 42.709 i + \cos 42.709 j) = 0.7348\alpha_{A/B} i + 0.6783\alpha_{A/B} j$$

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Example 4.2

The acceleration of point A given the velocity of B is

$$a_A = a_B + a_{A/B} + 2\omega_{AB} \times v_{A/B} + \alpha_{BA} \times r_{A/B} - \omega^2 r_{A/B}$$

$$a_A = 0 + 0.7348\alpha_{A/B} i + 0.6783\alpha_{A/B} j + 2\begin{bmatrix} i & j & k \\ 0 & 0 & 2 \\ -2.5907 & -2.3915 & 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & 10 \\ 1.3 & 1.2 & 0 \end{bmatrix} - (10)^2 (1.3i + 1.2j)$$

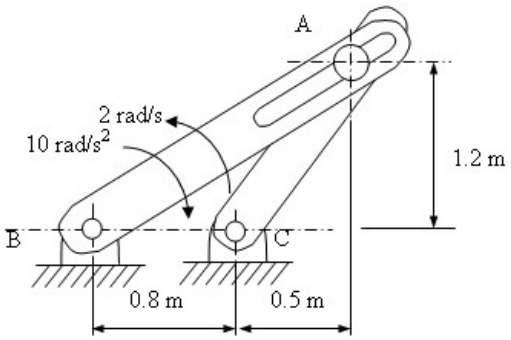
$$a_A = (0.7348\alpha_{A/B} - 137.217)i + (0.6783\alpha_{A/B} - 112.1815)j \quad (8)$$

Acceleration of A given the velocity of C is

$$a_A = a_C + \alpha_{CA} \times r_{A/C} - \omega^2 r_{A/C}$$

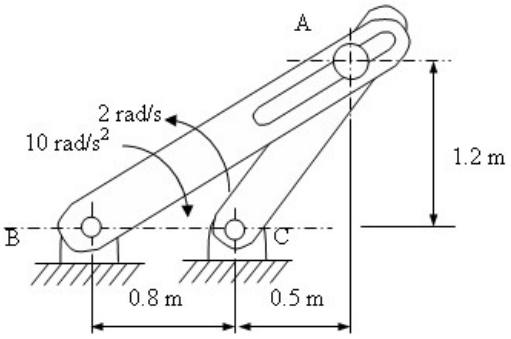
$$a_A = 0 + \begin{bmatrix} i & j & k \\ 0 & 0 & \alpha_{CA} \\ 0.5 & 1.2 & 0 \end{bmatrix} - 2^2 (0.5i + 1.2j)$$

$$a_A = -(1.2\alpha_{CA} + 2)i + (0.5\alpha_{CA} - 1.2j) \quad (9)$$



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Example 4.2



Equating i and j components of the two accelerations of point C yields ie Equations (8) and (9), we have

$$0.7348\alpha_{A/B} - 137.2171 = -1.2\alpha_{CA} + 2 \quad (10)$$

$$0.6783\alpha_{A/B} - 112.1815 = 0.5\alpha_{CA} - 1.2 \quad (11)$$

Solving the above two equations simultaneously, we have

$$\alpha_{A/B} = 171.658 \quad (\text{rad/s}^2)$$

$$a_{A//B} = 10.9017 \quad (\text{m/s}^2)$$

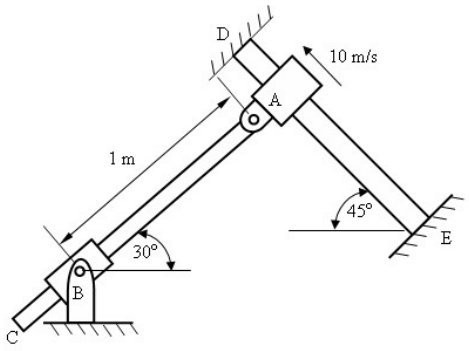
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Example 4.3

In Figure 1, the sleeve at A slides upward on the guide DE at a constant velocity of 10 m/s. The guide DE is inclined at 45° to the horizontal. Bar AC is pivoted to the sleeve at A with a pin and slides through the sleeve at B, which is pivoted to the ground. Using the vector analytical method, determine

a. the angular velocity and the sliding velocity of link AC,

b. the angular acceleration and the sliding acceleration of link AC



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Example 4.3 Solution

$v_A = v_B + v_{A/B} + \omega \times r_{A/B}$ (1)

$v_A = 10[cos(180 - 45)i + sin(180 - 45)j]$

$v_A = -7.071i + 7.071j$ (2)

Let $|v_{A/B}| = V$

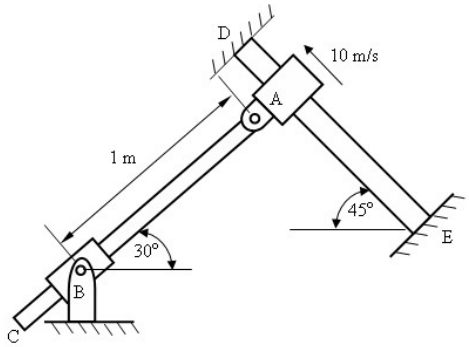
$v_{A/B} = V(\cos 30i + sin30j) = 0.866Vi + 0.5Vj$ (3)

$r_{A/B} = 1(\cos 30i + sin30j) = 0.866i + 0.5j$ (4)

Substituting (2) to (4) into (1), we have

$(-7.071i + 7.071j) = 0 + (0.866Vi + 0.5Vj) + \omega\, k \times (0.866i + 0.5j)$

$(-7.071i + 7.071j) = (0.866V - 0.5\omega)i + (0.5V + 0.866\omega)j$



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Example 4.3 Solution

Equating like terms of the above equation, we have

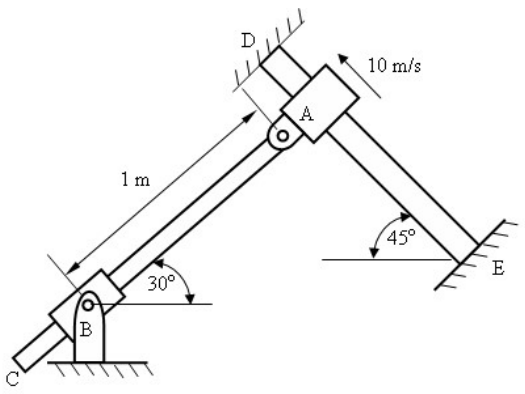
For i: $-7.071 = 0.866V - 0.5\omega$ (1)

j: $7.071 = 0.5V + 0.866\omega$ (2)

Solving the above equations simultaneously, we have

$$V = -2.5881\frac{m}{s}$$

$$\omega = 9.6594\, rad/s$$



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Example 4.3 Solution

$$a_A = a_B + a_{A/B} + 2\omega \times v_{A/B} + \alpha \times r_{A/B} - \omega^2 r_{A/B}$$

$$a_A = a_B = 0$$

Let $|a_{A/B}| = A$ Then $a_{A/B} = A(\cos 30i + sin30j)$

$$a_{A/B} = 0.866Ai + 0.5Aj$$

$$v_{A/B} = -2.5881\ (\cos 30i + sin30j)$$

$$v_{A/B} = -2.2413i - 1.2940j$$

$$r_{A/B} = 0.866i + 0.5j$$

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Example 4.3 Solution

$$0 = 0 + [0.866Ai + 0.5Aj] + 2\text{x}[9.6594k] \times [-2.2413i - 1.2940j] + [\alpha k] \times [0.866i + 0.5j] - (9.6594)^2[0.866i + 0.5j]$$

$$0 = (0.866A - 0.5\alpha - 55.8043i)i + (0.5A + 0.866\alpha - 89.9513)j$$

Equating like terms of the above equation, we have

For i: $55.8043 = 0.866A - 0.5\alpha$ (1)

j: $89.9513 = 0.5A + 0.866\alpha$ (2)

Solving the above equations simultaneously, we have

$$A = 93.3\,\frac{m}{s^2} \qquad \alpha = 9.6594\,rad/s^2$$

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